

Alex Mercer

Senior Software Engineer – Agentic AI & Cloud-Native Systems

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PROFILE

Senior Software Engineer with 12+ years of experience designing and delivering large-scale distributed systems, now specializing in **agentic AI architectures, LLM-driven automation, and cloud-native AI platforms**. Expert in orchestrating multi-agent workflows using frameworks such as **LangGraph, CrewAI, and custom Python-based agent runtimes**. Strong background in full-stack engineering, microservices, and high-performance backend systems. Passionate about building autonomous AI systems that reason, plan, and execute reliably in enterprise environments.

CORE COMPETENCIES

- **Agentic AI Systems:** Multi-agent orchestration, tool-use agents, planning agents, autonomous pipelines
- **LLM Engineering:** Prompt engineering, function calling, structured output, evaluation frameworks
- **RAG & Search:** Vector databases (Pinecone, Chroma, Vertex Matching Engine), retrieval optimization, hybrid search
- **Cloud Platforms:** GCP (Vertex AI, Cloud Run, Pub/Sub), AWS (SageMaker, Lambda), Azure OpenAI
- **Backend Engineering:** Python, Go, C#, Node.js, microservices, distributed systems
- **DevOps & MLOps:** Docker, Kubernetes, CI/CD, feature stores, model deployment
- **Data Engineering:** ETL pipelines, streaming systems, schema design, data governance

PROFESSIONAL EXPERIENCE

Senior Software Engineer – AI Systems

NovaAI Technologies, Toronto, ON 2021 – Present

- Designed and implemented a **multi-agent orchestration platform** enabling autonomous research, summarization, and decision-support workflows for enterprise clients.
- Built a **RAG pipeline** using Vertex AI embeddings + Pinecone, reducing hallucinations by 42% and improving retrieval precision.
- Developed a **task-planning agent** capable of decomposing complex business workflows into executable subtasks using LLM-based reasoning.
- Integrated tool-use capabilities enabling agents to call APIs, query databases, and trigger cloud functions.
- Led migration of legacy ML services to **containerized, serverless GCP architecture**, improving scalability and reducing costs by 28%.
- Mentored a team of 5 engineers on LLM engineering, prompt design, and agentic system patterns.

Software Engineer – Cloud & Distributed Systems

BlueWave Digital, Vancouver, BC 2016 – 2021

- Built high-throughput backend services in Go and Python supporting millions of daily transactions.
- Designed microservices architecture with event-driven patterns using Kafka and gRPC.
- Implemented automated CI/CD pipelines and containerized deployments on Kubernetes.
- Collaborated with data science teams to integrate ML inference endpoints into production systems.

Software Developer

NorthPeak Solutions, Calgary, AB 2013 – 2016

- Developed full-stack web applications using C#, .NET, and React.
- Built internal APIs and authentication systems for enterprise clients.
- Improved system reliability by implementing monitoring, logging, and automated testing frameworks.

PROJECT HIGHLIGHTS

- **Autonomous Research Agent** Built a multi-agent system that autonomously gathers data, evaluates sources, and produces structured reports for financial analysts.
- **AI-Driven Customer Support Assistant** Designed an agentic workflow combining retrieval, reasoning, and tool-use to automate 60% of Tier-1 support tasks.
- **Enterprise RAG Platform** Created a modular RAG architecture with pluggable retrievers, ranking models, and evaluation metrics.

EDUCATION

B.Sc. Computer Science University of British Columbia

CERTIFICATIONS

- Google Cloud Professional Cloud Developer
- AWS Certified Solutions Architect – Associate
- DeepLearning.AI: LLMs & LangChain Specialization

TECH STACK

Languages: Python, Go, C#, JavaScript/TypeScript **AI/ML:** LangGraph, LangChain, CrewAI, HuggingFace, Vertex AI, OpenAI APIs **Databases:** PostgreSQL, Redis, MongoDB, Pinecone, Chroma **Cloud:** GCP, AWS, Azure **Tools:** Docker, Kubernetes, Terraform, GitHub Actions